

DR. CHIA MIN YAN

Data Scientist | ML Engineer | Engineering PhD

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PROFILE SUMMARY

Dr. Chia Min Yan is an engineering PhD and Data Scientist working at the intersection of Machine Learning, Generative AI, and process optimization. He has published 5+ research articles in top-tier Q1 Web of Science journals. He designs and implements production-grade predictive models, scalable microservice architectures, and autonomous agentic workflows that bridge scientific engineering with commercial AI applications.

DEVELOPMENT & CORE TOOLS

Python (ML)	PyTorch
TypeScript	JavaScript
Node.js	Fastify
FastAPI	Angular
SQL	PostgreSQL
MATLAB	

DEVELOPER TOOLS

Docker	Kubernetes
Git & CI/CD	LangChain
LLMs (RAG)	

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

July 2026 – Present

Juice Up EV Grid Sdn. Bhd.

- Algorithms & Systems: Lead the end-to-end research, architecture, and deployment of production-grade statistical algorithms and systems (rewards program optimization, EV fleet charging schedules, station utilization) to drive EV charging network operations.
- Full Stack & Backend Architecture: Spearhead the technical design of scalable backend microservices, rewards database structures, and RESTful APIs, while directing frontend development of responsive EV fleet management portals and interactive customer dashboards.
- Generative AI Initiatives: Champion corporate AI strategies, orchestrating custom LLM integrations and advanced agentic frameworks to automate complex reasoning and EV customer support workflows.

Technologies: Python, TypeScript, Node.js, Angular, PostgreSQL, Git, Docker

Data Scientist

July 2022 – July 2026

PAIDChain Sdn. Bhd.

- Machine Learning Engineering: Research, build, and deploy core enterprise statistical models including real-time fraud detection systems, merchant risk anomaly trackers, system monitors, and algorithmic credit scoring structures.
- Full Stack Architecture: Lead the design and implementation of highly scalable backend RESTful APIs, database structures, and responsive web portals/interactive dashboards using Node.js, TypeScript, PostgreSQL, and Angular.
- Generative AI Initiatives: Champion corporate AI strategies by setting up, orchestrating, and launching custom LLM integrations, retrieval-augmented generation (RAG) models, and advanced agentic frameworks for workflow automation.

Technologies: Python, TypeScript, Node.js, Angular, PostgreSQL, Git, Docker, LangChain

Production Engineer

Jan 2019 – April 2019

Maxter Glove Manufacturing Sdn. Bhd.

- Monitored glove quality assurance workflows and parameters to satisfy international regulatory compliance and client product specifications.
- Standardized preventative line maintenance records and engineered key operations tracking documents to minimize system downtime.

Production Engineer (Intern)

Oct 2015 – Dec 2015

Maxter Glove Manufacturing Sdn. Bhd.

- Supervised and maintained treatment variables for the industrial wastewater facility, guaranteeing 100% compliance with national ecological parameters.
- Designed chemical dosing optimization strategies, successfully lowering operation costs and reducing chemical waste.

LANGUAGES

- English — Professional
- Chinese (Mandarin) — Native
- Bahasa Melayu — Professional
- Korean — Conversational

RESEARCH INTERESTS

- Environmental System Modelling
- Agentic Decision Architectures
- Applied ML in Process Systems
- Polymer composite membranes
- Sustainable fuel cells

PROFESSIONAL AFFILIATIONS

- Graduate Member — Board of Engineers Malaysia (BEM)
- Graduate Member — Institution of Engineers, Malaysia (IEM)
- Associate Member — Inst. of Chemical Engineers (ICChemE)
- Graduate Technologist — Malaysia Board of Tech (MBOT)

EXTRACURRICULARS

- UTAR Chinese Debate Team (2018–2022)
- Completed in the 16th and 17th National Varsity Chinese Debate Tournaments.
- Represented UTAR in the 2018 Nan Ying Cup Debate in China.

EDUCATION

Doctor of Philosophy (Engineering) 2019 – 2022
Universiti Tunku Abdul Rahman, Malaysia
Thesis: Robust Data Fusion Techniques Integrated Machine Learning Models for Estimating Reference Evapotranspiration. Developed deep learning models and custom stochastic optimization algorithms to resolve data sparsity.

Master of Engineering Science 2017 – 2019
Universiti Tunku Abdul Rahman, Malaysia
Dissertation: Development of High Performance Sulfonated Poly (Ether Ether Ketone) Based Composite Proton Exchange Membranes. Investigated chemical parameters to increase proton fuel cell membrane performance.

Bachelor of Engineering (Honours) Chemical Engineering 2013 – 2016
Universiti Tunku Abdul Rahman, Malaysia
CGPA: 3.8758 / 4.0000 (First Class Honours with Distinction)

SELECTED PUBLICATIONS

Wong, Y.J.*, Yeganeh, A., Chia, M.Y.*, et al. 2023. Quantification of COVID-19 impacts on NO2 and O3: Systematic model selection and hyperparameter optimization on AI-based meteorological-normalization methods. Atmospheric Environment. [Q1]

Wai, K.P., Chia, M.Y., Koo, C.H.*, Huang, Y.F. and Chong, W.C. 2022. Applications of deep learning in water quality management: A state-of-the-art review. Journal of Hydrology. [Q1]

Chia, M.Y., Huang, Y.F.*, Koo, C.H., Ng, J.L., Ahmed, A.N. and El-Shafie, A. 2022. Long-term forecasting of monthly mean reference evapotranspiration using deep neural network: A comparison of training strategies and approaches. Applied Soft Computing. [Q1]

Chia, M.Y., Huang, Y.F.* and Koo, C.H. 2022. Resolving data-hungry nature of machine learning reference evapotranspiration estimating models using inter-model ensembles with various data management schemes. Agricultural Water Management. [Q1]

Chia, M.Y., Huang, Y.F.* and Koo, C.H. 2021. Improving reference evapotranspiration estimation using novel inter-model ensemble approaches. Computers and Electronics in Agriculture. [Q1]

REFERENCES AVAILABLE UPON REQUEST